alt.fun: A Launchpad Backed by Hyperliquid Perps

alt.fun
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Abstract

The system described here is a token launchpad in which the reserve asset of every bonding curve is a tokenized leveraged perpetual position rather than a spot base asset. Existing launchpads produce momentum vehicles: tokens whose price moves only when someone trades them. Our design produces hybrid assets, called *alt coins* (abstracted leveraged tokens): tokens that are simultaneously momentum vehicles and directional bets on a chosen underlying, packaged in a single ERC-20. An alt coin's price evolves continuously with the underlying, even when no one is trading. The set of available underlyings is the set of Hyperliquid perpetuals, currently spanning crypto, equities, commodities, and indices via HIP-3. This paper presents the bonding curve, the graduation flow that migrates liquidity to a constant-product AMM at the last curve price, and the properties that follow from the design.

1. Introduction

Token launchpads have converged on a common design. A token with fixed supply is priced by a bonding curve, denominated in a spot base asset. Trades against the curve move the token's price relative to that base asset. After a threshold is reached, liquidity migrates to an AMM and trading continues there.

This paper specifies a launchpad design in which the base asset is replaced with a tokenized leveraged perpetual position. The resulting hybrid assets are *alt coins*: tokens that combine the social trading dynamics of a launchpad token with the directional exposure of a leveraged position, in a single ERC-20.

An alt coin's price moves from two sources: trading on the curve, and movement of the underlying amplified by the chosen leverage. The alt coin continues to evolve in price even when no one is trading. The set of viable underlyings is the set of Hyperliquid [1] perpetuals, which through HIP-3 [2] now spans crypto, equities, commodities, and indices. Creators choose the underlying, the direction (long or short), and the leverage factor (2x, 3x, or 5x) at launch.

The constant-product pricing math is preserved, but the system around it must adapt to a reserve that moves on its own.

2. The Bonding Curve

Each launched token is priced by a constant-product bonding curve, adapted from the Virtuals Protocol bonding curve implementation [3]. The reserves are the token itself and a tokenized leveraged position on a Hyperliquid perpetual (the *leveraged token*, or LT, issued by BounceTech [4]). The pair maintains the invariant

$$x \cdot y = k$$

where x is the token reserve, y is the LT reserve, and k is determined at launch.

The curve uses virtual reserves. At launch, the pair is initialized with

$$x_0 = S, \quad y_0 = \frac{V}{p_{LT}}, \quad k = x_0 \cdot y_0$$

where S is the total supply (one billion units), V is a fixed USD seed value (currently \$3,000), and p_{LT} is the LT exchange rate at launch. The pair's actual ERC-20 balances at launch are 75% of S (the *curve reserve*) and zero LT. The remaining 25% (the *LP reserve*) is held outside the curve and used at graduation; its mechanics are described in Section 3.

This setup produces three properties.

Deterministic supply trigger. Because the pair holds only 75% of the token supply as real reserve, the latest possible buy drains the token reserve to zero, at which point all 750M curve tokens have been sold. This is one of the two graduation triggers (Section 3).

Constant opening market cap. Token price in USD is

$$p_{token} = \frac{y \cdot p_{LT}}{x}$$

At launch this evaluates to $p_{token,0} = (y_0 \cdot p_{LT})/x_0 = V/S$. The opening market cap is therefore

$$\mathrm{MC}_0 = p_{token,0} \cdot S = V$$

regardless of which LT a creator pairs with. Every token opens at the same dollar valuation. The curve's slope adapts to the LT's price.

Two sources of price evolution. The token price expression $p_{token} = y \cdot p_{LT}/x$ is a function of two independent variables: the reserve ratio y/x, which moves only when trades occur on the curve, and the LT exchange rate p_{LT}, which moves with the underlying perpetual's price amplified by the LT's leverage. A token's USD price evolves continuously with the underlying even when no trades occur.

3. Graduation

A token graduates from the bonding curve to a constant-product AMM [5]. Graduation is the migration of the curve's accumulated liquidity into a standard ERC-20 trading pair. After graduation, the bonding curve is closed and all subsequent trading happens on the AMM.

Triggers

A token graduates when either condition is met:

USD trigger. The USD value of the LT held in the curve reaches a threshold T:

$$y \cdot p_{LT} \geq T$$

where y is the real LT balance in the pair and p_{LT} is the LT exchange rate. The default threshold is $9,000.

Supply trigger. The token's real curve reserve is exhausted:

$$x_{\text{real}} = 0$$

i.e. all 750M curve tokens have been sold.

The two triggers cover qualitatively different paths to graduation. The USD trigger captures cases where the underlying rallies sharply and the LT's appreciation pushes the curve's USD value over the threshold even with limited buy activity. The supply trigger captures cases where the underlying is flat or declining and the curve sells out without the LT ever reaching the USD threshold.

Either trigger routes through the same migration logic.

Migrating to the AMM at the last curve price

The migration must seed the AMM such that the opening AMM price equals the last curve price. Let r_0 and r_1 be the pair's virtual reserves immediately before graduation, and ℓ be the real LT balance held in the pair at graduation. The amount t_{LP} of tokens that, paired with ℓ, opens the AMM at the same price as the curve is

$$t_{LP} = \ell \cdot \frac{r_0}{r_1}$$

This makes the AMM's reserve ratio identical to the curve's last reserve ratio, so the marginal price is preserved across the transition. There is no arbitrage gap for searchers to capture at the moment of graduation.

Sufficiency of the LP reserve

The 25% LP reserve held by the protocol at launch must be enough to seed the AMM under any feasible graduation. Let s be the real number of tokens sold from the curve before graduation. The curve mechanics imply that the LT balance ℓ and the virtual reserves r_0, r_1 at graduation are both functions of s. Substituting into the expression above gives

$$t_{LP}(s) = \frac{s \cdot (S - s)}{S}$$

where S is the total supply. This is a parabola in s with maximum at $s = S/2$, where $t_{LP}(S/2) = S/4$. This is exactly 25% of supply, which is exactly the LP reserve (Figure 1).

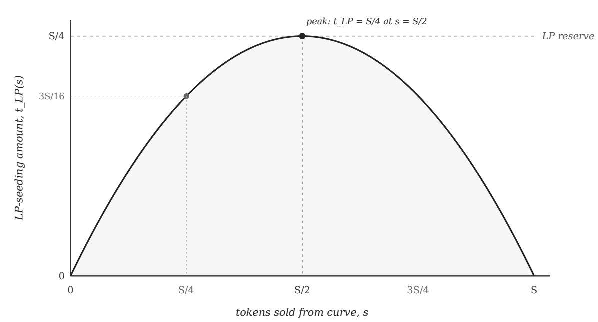

Figure 1: The required LP-seeding amount $t_{LP}(s)$ as a function of tokens sold from the curve s. The parabola peaks at $s = S/2$ with value $S/4$, exactly matching the protocol's held LP reserve.

The protocol therefore holds, by construction, the maximum amount of tokens needed for graduation under any path. When $s \neq S/2$, the required t_{LP} is less than $S/4$, and the surplus is burned. The LP reserve is sufficient by mathematical guarantee, not by parameter choice.

Permissionless graduation

The transaction that completes graduation can be submitted by anyone, not only the protocol operator. The system maintains a backstop keeper for normal operation, but the keeper is not a privileged actor: any token in the graduating state can be graduated by any sender. The protocol cannot withhold a token from graduation once the trigger condition has been met.

4. Properties

The design produces several properties that follow directly from the bonding curve and graduation mechanics described above.

Asset universe

A token's underlying is a Hyperliquid perpetual that has been wrapped as a leveraged token by BounceTech, the LT issuer. Viable underlyings are therefore those perpetuals for which BounceTech has issued an LT. As BounceTech adds support for additional Hyperliquid markets, those markets become available for new launches.

A token's underlying is fixed at launch in the current implementation. Future versions may admit reassignment of the underlying through governance.

Hybrid assets

A token launched on this system represents two positions in a single ERC-20: a position in the token itself, and a leveraged position in the underlying perpetual that backs the reserve. Buying the token is buying both. Selling the token is selling both.

This combination has no clean precedent. Existing launchpads produce momentum vehicles. Existing perpetuals are positions held by individual traders against a counterparty. The design here produces an instrument that carries the social properties of a launched token (a public ticker, a community, organic distribution) and the financial properties of a leveraged directional bet (continuous mark-to-market, exposure to a chosen underlying).

Continuous price evolution

A token's USD price is a function of two independent variables. The reserve ratio y/x updates only when trades occur on the curve. The LT exchange rate p_{LT} updates with every change in the underlying perpetual's price. A token therefore re-prices in USD whenever the underlying moves, regardless of trading activity. A token can graduate, appreciate, or depreciate during periods in which no trades occur.

Anti-snipe trading delay

Trading on a newly launched token is gated for a short window after deployment. The mechanism is a per-token delay enforced via EIP-1153 transient storage: trades attempted before the delay elapses revert. The delay is currently three blocks. The window narrows the asymmetry between sniping bots and human users by absorbing the first opportunity for atomic same-block extraction.

Identifying address pattern

Every alt.fun token is deployed at an address ending in a fixed suffix. The launch contract enforces this on-chain by rejecting any deployment whose resulting address does not match the suffix; matching addresses are produced off-chain by mining a CREATE2 salt. The suffix lets indexers, aggregators, and frontends positively identify alt.fun tokens at the address level alone, without needing a registry lookup. Tokens that copy the ticker, image, or metadata of an alt.fun token but are deployed elsewhere will not match the pattern and can be filtered structurally.

The protocol enforces a minimum suffix length. Creators may mine longer suffixes voluntarily; matching each additional character is exponentially more proof-of-work.

Permanent LP lock

The LP tokens minted at graduation are sent to a custodian contract with no withdraw function. The liquidity is permanently locked. Neither the protocol nor the creator can extract it, and there is no governance path to recover it in the current implementation.

5. Risks

Volatility decay

alt.fun tokens are paired with leveraged tokens issued by BounceTech. A leveraged perpetual position would be liquidated if its margin fell below the maintenance threshold. BounceTech's LTs avoid liquidation by rebalancing the underlying perp position as price moves: the position is shrunk after losses to restore margin, and grown after gains to maintain target leverage. Rebalancing keeps the position alive but converts realized volatility into a drag on net asset value. In oscillating markets without a clear directional trend, the LT loses value even when the underlying ends at the same price it started.

This is not specific to any LT issuer or rebalancing schedule. Any constant-leverage instrument that avoids liquidation must rebalance, and any rebalancing strategy decays in choppy conditions. Decay in exchange for the absence of liquidation risk is intrinsic to the asset class.

Higher leverage amplifies the effect. A token paired with a 5x LT decays faster in choppy conditions than one paired with a 2x LT.

NAV approaching zero

Sustained adverse moves in the underlying can push the LT's net asset value toward zero. The LT does not liquidate because rebalancing absorbs adverse moves by reducing the position size, but the asymptotic floor is zero. A token paired with a 5x long LT on an underlying that drops sharply can see its reserve approach zero, which directly reduces the USD value of every holder's position. The effect grows with leverage.

Bidirectional graduation distance

A token approaching the graduation threshold can be pushed back by movement in the underlying against the position's direction. A token at 80% of the USD threshold can return to 60% on a single bad session for the underlying. Graduation is not a one-way ratchet. Holders positioning for imminent graduation accept the risk that graduation moves further away.

Infrastructure dependencies

The system depends on external infrastructure: Hyperliquid for perpetual execution and order matching, and the LT issuer for atomic mint and redeem. Downtime, exploits, or unexpected behavior in either disrupts the system. The design does not insulate against failure of either dependency.

Redemption capacity

LT redemptions are atomic and capped by the LT's idle USDC buffer. In ordinary conditions this is invisible: redemptions clear immediately. Under mass-sell pressure, the atomic redemption buffer can be temporarily depleted and individual sell transactions revert until the buffer is automatically replenished, which requires bridging funds from HyperCore to HyperEVM. Sellers face execution risk during these periods. The risk is a property of the atomic-only redemption design; an asynchronous queue would absorb the pressure but introduce settlement delay on every trade. After graduation, sellers can route around buffer depletion by swapping their tokens for LT on the AMM pool directly.

6. Conclusion

This paper specifies a token launchpad in which the reserve asset of every bonding curve is a tokenized leveraged perpetual position. Tokens launched on this system are simultaneously momentum vehicles and directional bets on a chosen underlying. They re-price continuously with the underlying, can graduate from market movement alone, and migrate to a constant-product AMM at the last curve price by mathematical guarantee.

References

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